

NEWT'S AND RAPH'S

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Project Overview

This project aims to implement the Newton-Raphson Method for Optimization [1] polynomial function $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$, in matrix form:

$$P_{f(x)} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ \vdots \end{bmatrix}$$

It does so by multiplying it with a matrix constructed from the size of the polynomial in matrix form, the matrix for the derivative operation constructed being:

$$D_{\text{matrix}} = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots \\ & 0 & 2 & 0 & \dots \\ & & 0 & 3 & \dots \\ & & & 0 & \ddots \end{bmatrix}$$

Figure 1: The derivative operation for polynomials, represented as a derivative.

By multiplying them both, we get $f'(x)$ [2]:

$$P_{f'(x)} = D_{\text{matrix}} \times P_{f(x)}$$

Evaluating the polynomial at a given point is simple, by raising x elementwise to the index of the polynomial in matrix form (the degree of each of the terms), then multiplying it to the coefficient defined in the polynomial matrix form.

$$f(x) = \sum_{i=0}^n x^{\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \\ n \end{bmatrix}} \circ \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{bmatrix}$$

$$\text{where } x^{\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \\ n \end{bmatrix}}$$

is the elementwise operation of raising x to every element from 1 to n

Objectives

- Develop a library that can calculate the Newton-Raphson method and other prerequisites operations (differentiation and polynomial evaluation).
- Provide a simple user interface that can expose the functionality of the library to a non-command line user.
- Check if the library correctly calculates the optimal values of a given function, up to tolerance.

Showcase

Newton-Raphson Method for Optimization

Add or delete additional terms:

count = 4

c

x

x^2

x^3

x^4

Figure 2: The title bar, plus the inputs for each of the terms of the polynomial.

Polynomial

$$f(x) = -3.0 - 5.0x - 7.0x^2 + 2.0x^3 + 4.0x^4$$

Derivative

$$f'(x) = -5.0 - 14.0x + 6.0x^2 + 16.0x^3$$

Second Derivative

$$f''(x) = -14.0 + 12.0x + 48.0x^2$$

Figure 3: Show the input function, typeset with LaTeX.

0
Initial Guess
0.05
Tolerance

Figure 4: Enter initial guess and tolerance.

Newton-Raphson Method for Optimization

Polynomial: $f(x) = -3.0 - 5.0x - 7.0x^2 + 2.0x^3 + 4.0x^4$

Initial Guess: $x_1 = 0.0$

Tolerance: $\epsilon = 0.05$

Optimal value: -0.35714285714285715

Optimal solution: -2.133173677634319

Figure 5: Display results of Newton-Raphson method for optimization



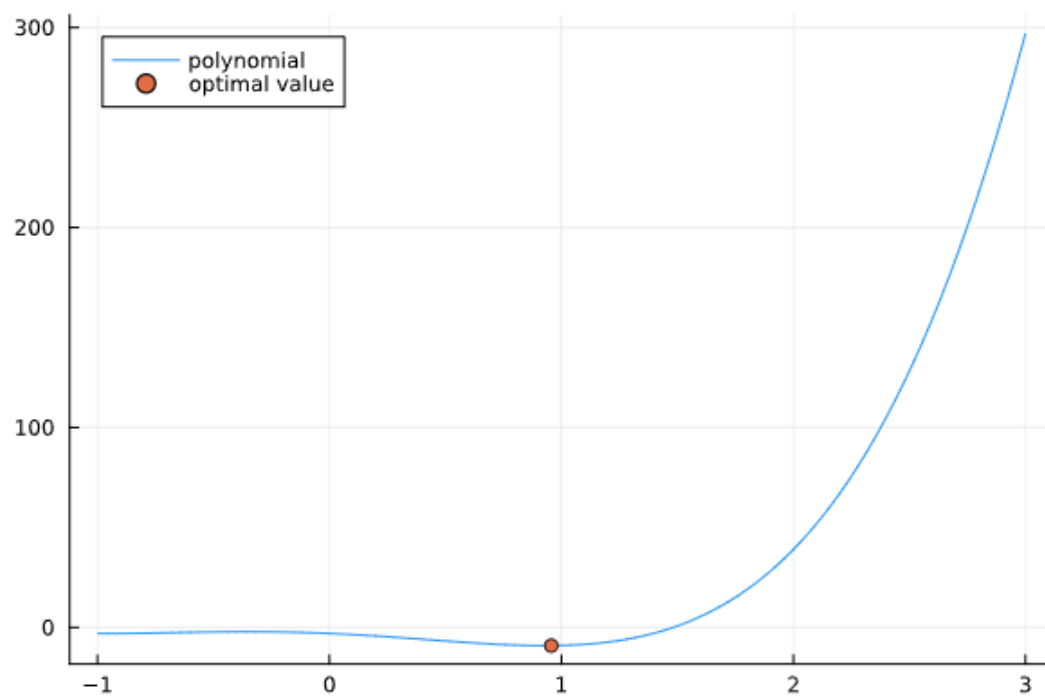
Display table of optimization



Plot optimal values

Min

Max



4x2 DataFrame

Row	x_i Float64	ε Bool
1	2.03304	false
2	1.43059	false
3	1.09435	false
4	0.955976	true

Figure 6: Display graph and table.

Recommendations

- Extend the solver to be able to calculate derivatives for trigonometric and exponential functions.
- Implement an os-native user interface.

Citations

- [1] "Newton's method in optimization." Accessed: Dec. 15, 2025. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Newton%27s_method_in_optimization&oldid=1311976880
- [2] "Derivatives via matrix multiplication." [Online]. Available: <https://youtu.be/SSPbgRQCqUo>